Power System Fault Analysis

Introduction to Faults in Power Systems

A fault is an abnormal condition in a power system that causes excessive current and/or abnormal voltages.

Common Causes

- Lightning and insulation failure
- Equipment failure or ageing
- Trees, animals, or foreign objects contacting conductors
- Mechanical damage and human error

Why Fault Analysis is Important

Faults can cause:

$$\boxed{\text{High Fault Current} \rightarrow \text{Equipment Damage} \rightarrow \text{System Disturbance}}$$

Fault analysis is required to:

- Determine the magnitude of fault current
- Select appropriate circuit breakers and protective relays
- Ensure safe and reliable operation of the power system

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Basic Classification

$$\boxed{\text{Power-System Faults} \rightarrow \begin{cases} \text{Symmetrical Faults} \\ \text{Unsymmetrical Faults} \end{cases}}$$

Symmetrical: $L-L-L,\ L-L-L-G$

Unsymmetrical: $L-G,\ L-L,\ L-L-G$

Defining the Shunt Fault

A shunt fault is an **abnormal low-resistance path** that bypasses the normal load circuit — arising from direct physical contact between conductors, or between a conductor and earth.

Low-Resistance Path

Unintended connection dramatically reduces impedance at the fault point

Extreme Current Surge

Current rises far beyond rated values, stressing all connected equipment

Physical Contact

Conductor–conductor or conductor–ground contact triggers the fault



Fault Classification

Symmetrical Faults

Three-Phase (LLL)

All three phases short together — balanced, analysed via single-phase equivalent

Three-Phase-to-Ground (LLLG)

All phases shorted to ground — most severe, positive-sequence network only

Unsymmetrical Faults

Single Line-to-Ground (LG)

Most common — sequence networks connected in series

Line-to-Line (LL)

Two phases contact — Z1 and Z2 networks in parallel

Double Line-to-Ground (LLG)

Two phases and ground — Z1 in series with (Z0 ‖ Z2)

Fault Calculations for Shunt Faults – Overall Procedure

Fault calculations are performed to determine the current flowing during a short-circuit condition.

The general procedure is:

1. Draw the one-line diagram of the power system.
2. Identify the fault location and type of fault.
3. Select common base MVA and base voltage.
4. Convert all system impedances to a common per-unit base.
5. Construct the required sequence networks.
6. Reduce the sequence networks to the fault point.
7. Connect the sequence networks according to the fault type.
8. Calculate sequence currents.
9. Convert sequence currents to phase currents.
10. Convert per-unit values to actual values, if required.

$$\boxed{\text{System Data} \rightarrow \text{PU System} \rightarrow \text{Sequence Networks} \rightarrow \text{Fault Current}}$$

Step 1 – One-Line Diagram and Common Base

Draw the one-line diagram showing:

- Generators
- Transformers
- Transmission lines
- Buses
- Fault location

Select:

$$S_B = \text{Common Base MVA}$$

and choose the base voltage at one bus.

Base voltages at other buses are determined using transformer voltage ratios.

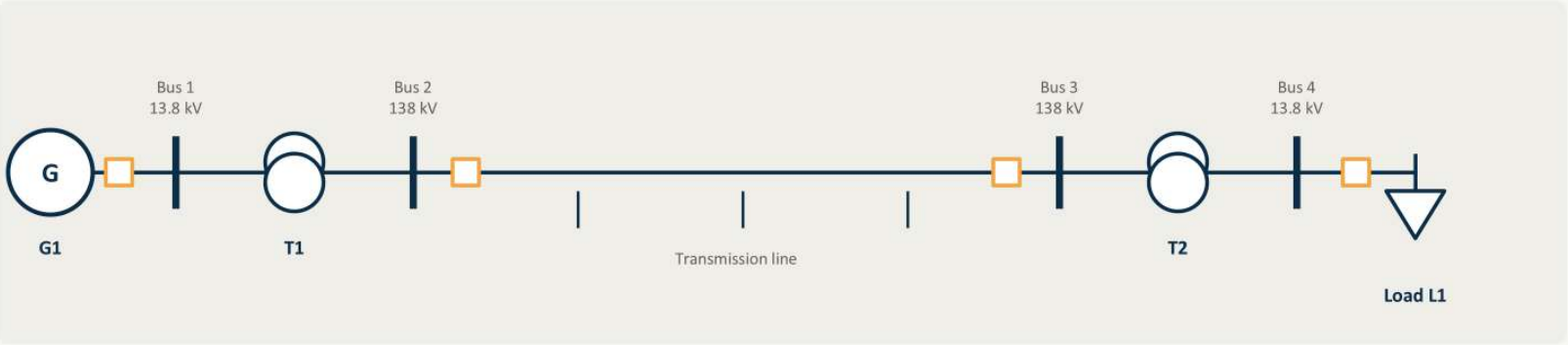
Convert all impedances to the selected base:

$$Z_{pu,new} = Z_{pu,old} \left(\frac{S_{B,new}}{S_{B,old}} \right) \left(\frac{V_{B,old}}{V_{B,new}} \right)^2$$

After this step, all system quantities are on a common per-unit base.

What is a one-line diagram?

A power system has three phases, but they're normally balanced and identical — so engineers draw just one phase as a single line, with a symbol standing in for each piece of equipment.



It's the map every fault, protection, and load-flow study starts from.

The symbols



Generator

Bus



Transformer



Circuit breaker



Transmission line



Load

Step 2 – Construct Sequence Networks

Construct the three sequence networks of the system:

Positive-sequence network

$$Z_1$$

- Represents normal phase sequence
- Contains the generator/source voltage

Negative-sequence network

$$Z_2$$

- Represents reverse phase sequence
- Normally contains no voltage source

Zero-sequence network

$$Z_0$$

- Represents in-phase components
- Depends strongly on transformer connections and grounding

Reduce each network as viewed from the fault point:

$$\boxed{Z_{1,eq}, \quad Z_{2,eq}, \quad Z_{0,eq}}$$

How Do Negative- and Zero-Sequence Currents Flow Without Sources?

A balanced generator produces only positive-sequence EMF:

$$\boxed{E_1 \neq 0, \qquad E_2 = 0, \qquad E_0 = 0}$$

During an unbalanced fault, the fault condition connects the sequence networks together.

For example, for a single line-to-ground (L-G) fault:

$$\boxed{E_1 \rightarrow Z_1 \rightarrow Z_2 \rightarrow Z_0}$$

The positive-sequence source E_1 drives current through all three sequence networks:

$$I_1 = I_2 = I_0 = \frac{E_1}{Z_1 + Z_2 + Z_0}$$

Therefore,

$$\boxed{E_2 = 0,\ E_0 = 0 \quad \text{but} \quad I_2 \neq 0,\ I_0 \neq 0}$$

The negative- and zero-sequence currents produce voltage drops across their respective impedances:

$$V_2 = -I_2 Z_2, \qquad V_0 = -I_0 Z_0$$

Step 3 – Connect Sequence Networks According to Fault Type

The sequence networks required depend on the type of shunt fault.

Fault Type	Sequence Networks
Three-phase fault (L-L-L)	Positive only
Single line-to-ground (L-G)	Positive + Negative + Zero
Line-to-line (L-L)	Positive + Negative
Double line-to-ground (L-L-G)	Positive + Negative + Zero

For example, for an L-G fault:

$$\boxed{I_0 = I_1 = I_2}$$

and the three sequence networks are connected in series.

For an L-L fault:

$$\boxed{I_0 = 0}$$

so the zero-sequence network is not involved.

Why Are All Sequence Currents Equal in an L-G Fault?

Consider a single line-to-ground fault on phase a.

At the fault point:

$$I_a = I_f, \qquad I_b = 0, \qquad I_c = 0$$

Use the phase-to-sequence transformation matrix:

$$\begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_f \\ 0 \\ 0 \end{bmatrix}$$

Multiplying,

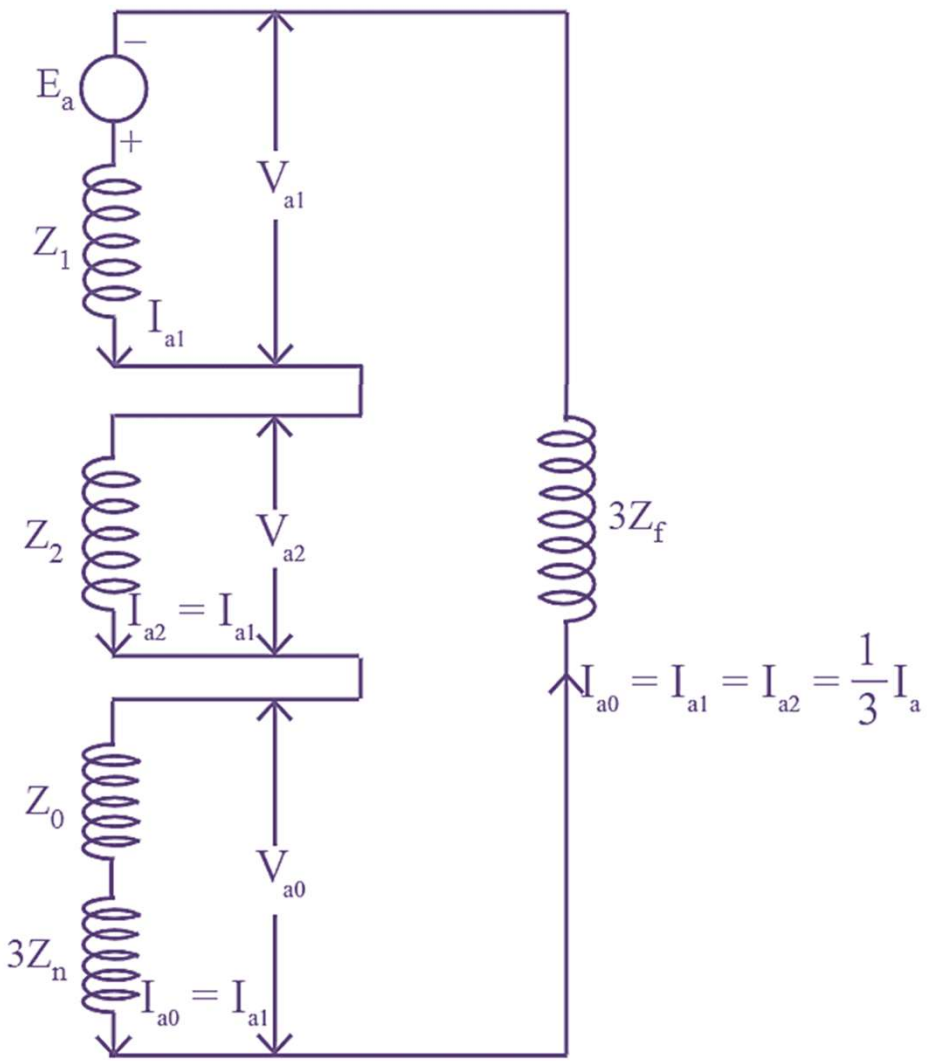
$$\begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} I_f \\ I_f \\ I_f \end{bmatrix}$$

Therefore,

$$\boxed{I_0 = I_1 = I_2 = \frac{I_f}{3}}$$

Hence, the same current must flow through all three sequence networks.

$$\boxed{I_0 = I_1 = I_2 \;\Rightarrow\; \text{Positive, Negative and Zero Sequence Networks in Series}}$$


E_a
−
+
Z_1
I_a1
V_a1
Z_2
I_a2 = I_a1
V_a2
3Z_f
I_a0 = I_a1 = I_a2 = 1/3 I_a
Z_0
V_a0
3Z_n
I_a0 = I_a1

Step 4 – Calculate Sequence Currents

After connecting the sequence networks, calculate:

$$I_0, \qquad I_1, \qquad I_2$$

The equations depend on the type of fault.

For example, for a solid L-G fault:

$$I_0 = I_1 = I_2 = \frac{E}{Z_1 + Z_2 + Z_0}$$

Hence the faulted phase current is:

$$I_a = I_0 + I_1 + I_2$$

$$\boxed{I_a = \frac{3E}{Z_1 + Z_2 + Z_0}}$$

Usually,

$$E \approx 1\angle 0^\circ \; pu$$

for basic fault calculations.

Step 5 – Convert Sequence Currents to Phase Currents

Once I_0, I_1, I_2 are known, obtain the actual phase quantities using the transformation matrix:

$$\boxed{\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix}}$$

where

$$a = 1\angle 120^\circ$$

This gives the fault currents in phases a, b, c.

Step 6 – Convert PU Current to Actual Current

If the calculated current is in per unit, determine the base current:

$$I_B = \frac{S_B}{\sqrt{3} V_B}$$

Then,

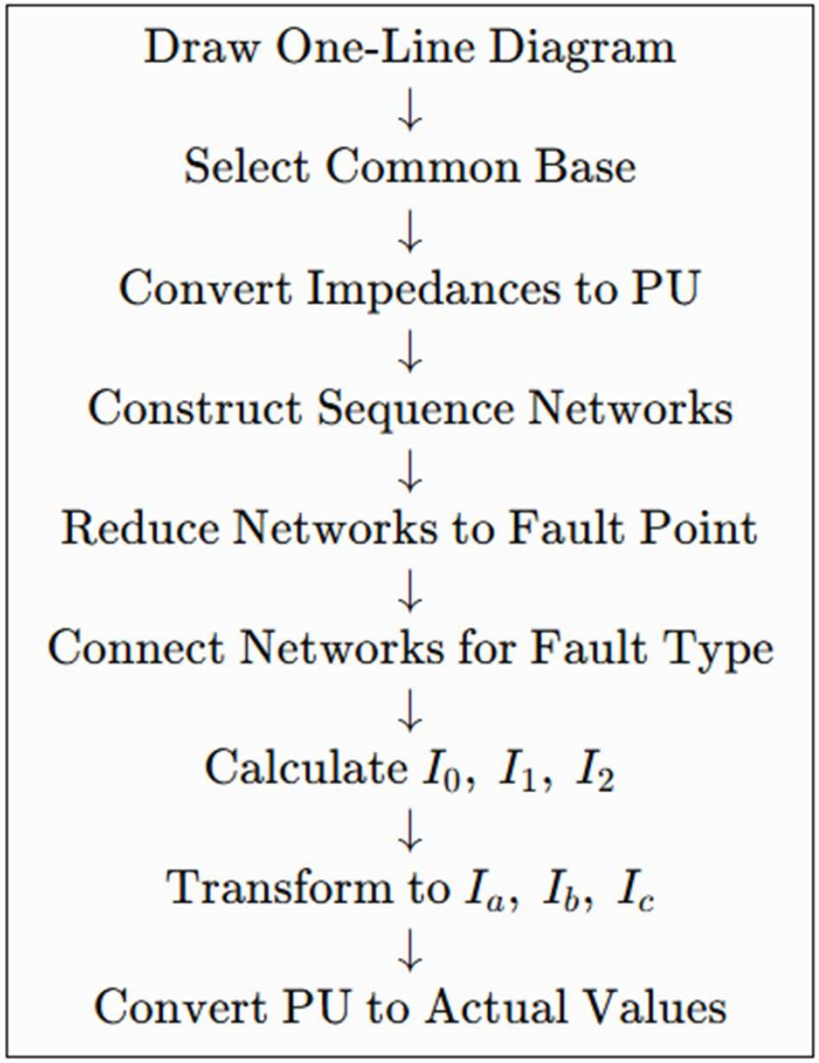
$$\boxed{I_{\text{actual}} = I_{pu} \times I_B}$$

For S_B in MVA and V_B in kV:

$$\boxed{I_B(\text{kA}) = \frac{S_B(\text{MVA})}{\sqrt{3}\, V_B(\text{kV})}}$$

Final results may include:

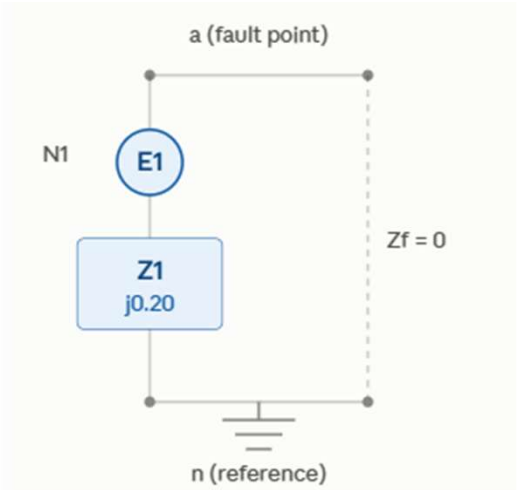
- Fault current in each phase
- Fault current in kA
- Sequence currents
- Bus voltages during the fault


Draw One-Line Diagram
Select Common Base
Convert Impedances to PU
Construct Sequence Networks
Reduce Networks to Fault Point
Connect Networks for Fault Type
Calculate $I_0,\ I_1,\ I_2$
Transform to $I_a,\ I_b,\ I_c$
Convert PU to Actual Values

A generator supplies a bus through a system having positive-sequence equivalent impedance

$$Z_1 = j0.20\ pu$$

A solid three-phase fault occurs at the bus. Determine the fault current.



For a symmetrical three-phase fault, only the positive-sequence network is involved.

$$I_1 = \frac{E_1}{Z_1}$$

Therefore,

$$I_1 = \frac{1}{j0.20}$$

Since

$$\frac{1}{j} = -j$$

$$I_1 = -j5 = 5\angle -90^\circ \; pu$$

For a balanced three-phase fault:

$$I_0 = 0, \qquad I_2 = 0$$

For a balanced three-phase fault:

$$I_0 = 0, \qquad I_2 = 0$$

Using the sequence-to-phase transformation:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} 0 \\ 5\angle -90^\circ \\ 0 \end{bmatrix}$$

Hence,

$$\boxed{I_a = 5\angle -90^\circ \; pu}$$

$$\boxed{I_b = 5\angle 150^\circ \; pu}$$

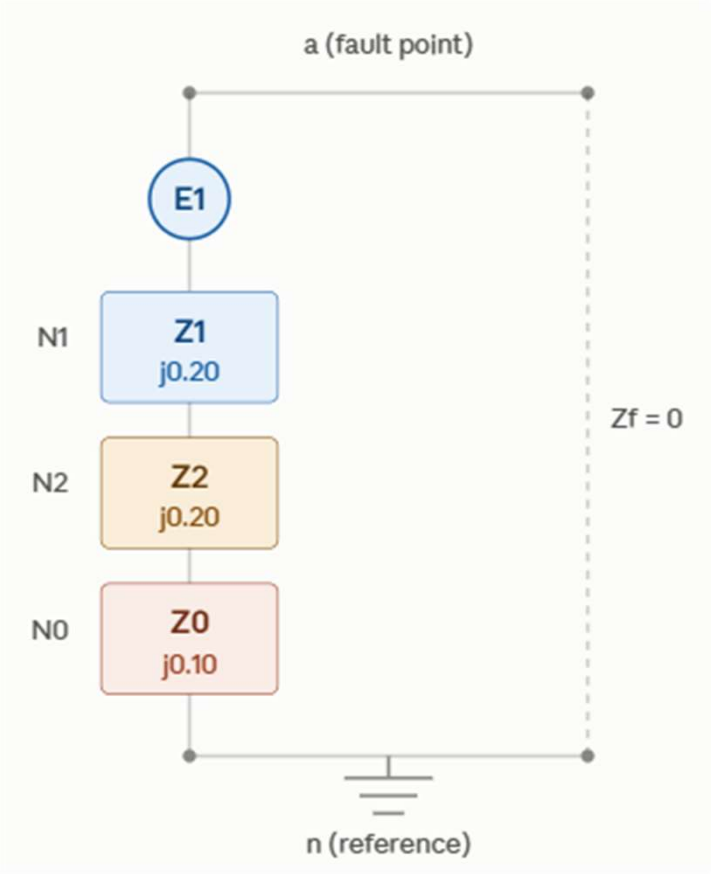
$$\boxed{I_c = 5\angle 30^\circ \; pu}$$

The three fault currents are equal in magnitude and 120° apart.

A three-phase fault occurs at an 11 kV bus. The system base is 100 MVA and the positive-sequence equivalent impedance is

$$Z_1 = j0.25\ pu$$

Determine the fault current in kA.



Fault current:

$$I_f = \frac{1}{j0.25}$$

$$\boxed{I_f = 4\angle -90^\circ \; pu}$$

Base current:

$$I_B = \frac{S_B}{\sqrt{3}V_B}$$

$$I_B = \frac{100}{\sqrt{3}(11)}$$

$$I_B = 5.249 \text{ kA}$$

Therefore,

$$I_{f,\text{actual}} = I_{f,pu} I_B$$

$$= 4(5.249)$$

$$\boxed{|I_f| = 21.0 \text{ kA}}$$

At a particular bus,

$$Z_1 = j0.20, \qquad Z_2 = j0.20, \qquad Z_0 = j0.10\ pu$$

A solid L-G fault occurs on phase a. Determine the sequence currents and phase currents.

Also determine the sequence currents and phase currents if there was a fault impedance Z_f =j0.05 pu

For an L-G fault:

$$I_0 = I_1 = I_2$$

The three sequence networks are connected in series.

Therefore,

$$I_1 = \frac{E_1}{Z_1 + Z_2 + Z_0}$$

$$I_1 = \frac{1}{j(0.20 + 0.20 + 0.10)}$$

$$I_1 = \frac{1}{j0.50} = -j2$$

Hence,

$$\boxed{I_0 = I_1 = I_2 = -j2\ pu}$$

Now use the transformation matrix:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} -j2 \\ -j2 \\ -j2 \end{bmatrix}$$

Therefore,

$$I_a = -j6$$

and because $1 + a + a^2 = 0$,

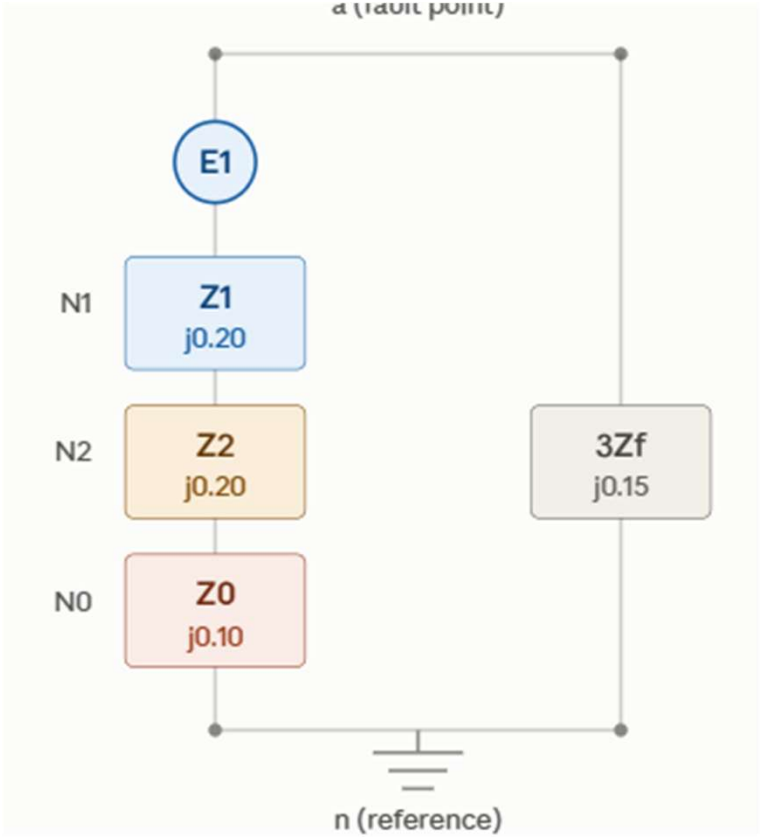
$$I_b = I_c = 0$$

Thus,

$$\boxed{I_a = 6\angle -90^\circ\ pu}$$

$$\boxed{I_b = I_c = 0}$$

This also verifies the physical condition of an L-G fault on phase a.



The fault impedance appears as $3Z_f$, therefore:

$$I_1 = \frac{E_1}{Z_1 + Z_2 + Z_0 + 3Z_f}$$

Substitute:

$$I_1 = \frac{1}{j0.20 + j0.20 + j0.10 + 3(j0.05)}$$

$$I_1 = \frac{1}{j0.65}$$

$$\boxed{I_1 = -j1.538\ pu}$$

Hence,

$$\boxed{I_0 = I_1 = I_2 = -j1.538\ pu}$$

The faulted phase current is:

$$I_a = I_0 + I_1 + I_2$$

$$I_a = 3(-j1.538)$$

$$\boxed{I_a = -j4.615\ pu}$$

or

$$\boxed{I_a = 4.615\angle -90^\circ\ pu}$$

A generator supplies a transmission line through a transformer. A solid line-to-line fault occurs between phases b and c at the receiving-end bus.

The sequence reactances on a common per-unit base are:

Generator:

$$X_{1G} = 0.15\ pu, \qquad X_{2G} = 0.20\ pu$$

Transformer:

$$X_{1T} = X_{2T} = 0.10\ pu$$

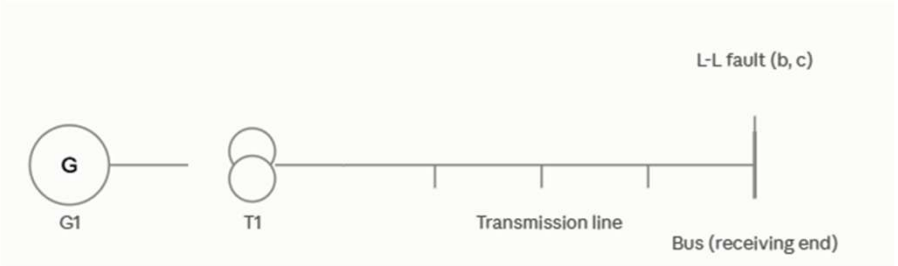
Transmission line:

$$X_{1L} = X_{2L} = 0.05\ pu$$

Assume

$$E_1 = 1\angle 0^\circ\ pu$$

Determine the sequence currents and phase fault currents.



Phase a is not involved in the fault:

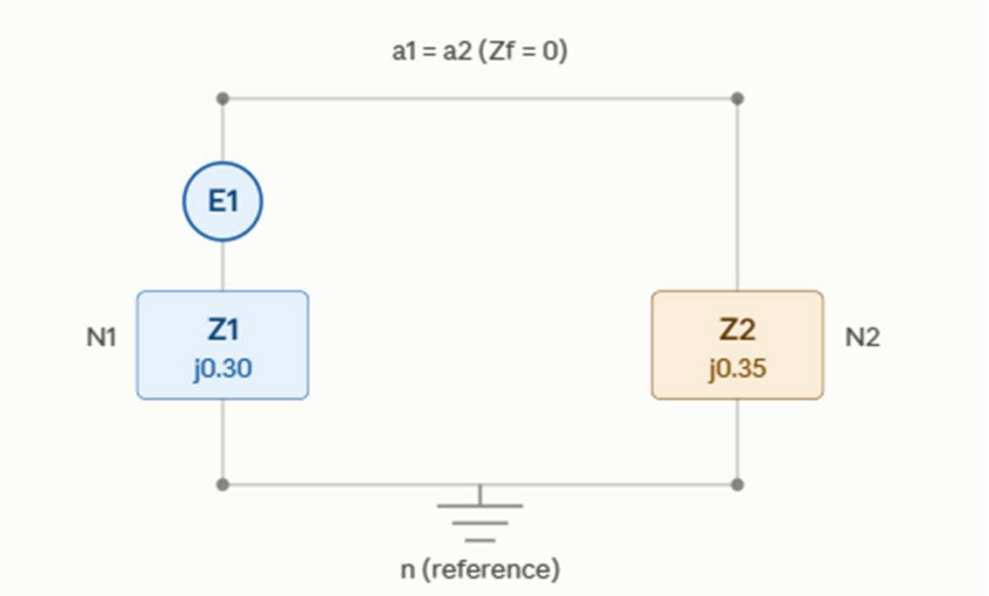
$$\boxed{I_a = 0}$$

There is no connection to ground, so the currents in the two faulted phases are equal and opposite:

$$\boxed{I_b = -I_c}$$

Also, because phases b and c are short-circuited together:

$$\boxed{V_b = V_c}$$



For a line-to-line fault:

$$I_0 = 0$$

and

$$I_2 = -I_1$$

First determine the positive-sequence equivalent impedance:

$$Z_1 = j(X_{1G} + X_{1T} + X_{1L})$$

$$Z_1 = j(0.15 + 0.10 + 0.05)$$

$$\boxed{Z_1 = j0.30\ pu}$$

Now determine the negative-sequence equivalent impedance:

$$Z_2 = j(X_{2G} + X_{2T} + X_{2L})$$

$$Z_2 = j(0.20 + 0.10 + 0.05)$$

$$\boxed{Z_2 = j0.35\ pu}$$

For a solid L-L fault,

$$I_1 = \frac{E_1}{Z_1 + Z_2}$$

Therefore,

$$I_1 = \frac{1}{j0.30 + j0.35}$$

$$I_1 = \frac{1}{j0.65}$$

$$\boxed{I_1 = -j1.538\ pu}$$

Hence,

$$\boxed{I_2 = j1.538\ pu}$$

and

$$\boxed{I_0 = 0}$$

Now obtain the phase currents using the transformation matrix:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} 0 \\ -j1.538 \\ j1.538 \end{bmatrix}$$

For phase a,

$$I_a = I_1 + I_2 = 0$$

$$\boxed{I_a = 0}$$

For phase b,

$$I_b = a^2 I_1 + a I_2$$

which gives

$$\boxed{I_b = -2.664\ pu}$$

For phase c,

$$I_c = a I_1 + a^2 I_2$$

which gives

$$\boxed{I_c = 2.664\ pu}$$

Therefore,

$$\boxed{I_a = 0, \qquad I_b = -2.664\ pu, \qquad I_c = 2.664\ pu}$$

As expected for a line-to-line fault,

$$\boxed{I_b = -I_c}$$

and no zero-sequence current flows:

$$\boxed{I_0 = 0}$$

At a bus,

$$Z_1 = j0.20, \qquad Z_2 = j0.20, \qquad Z_0 = j0.10\ pu$$

A solid L-L-G fault occurs on phases b and c.

Determine I_0, I_1, I_2.

At the fault point,

$$V_b = 0, \qquad V_c = 0$$

while phase a is not faulted:

$$I_a = 0$$

Now use the sequence-to-phase voltage transformation:

$$V_b = V_0 + a^2 V_1 + a V_2 = 0$$

$$V_c = V_0 + a V_1 + a^2 V_2 = 0$$

Subtracting the two equations,

$$(a^2 - a)(V_1 - V_2) = 0$$

therefore,

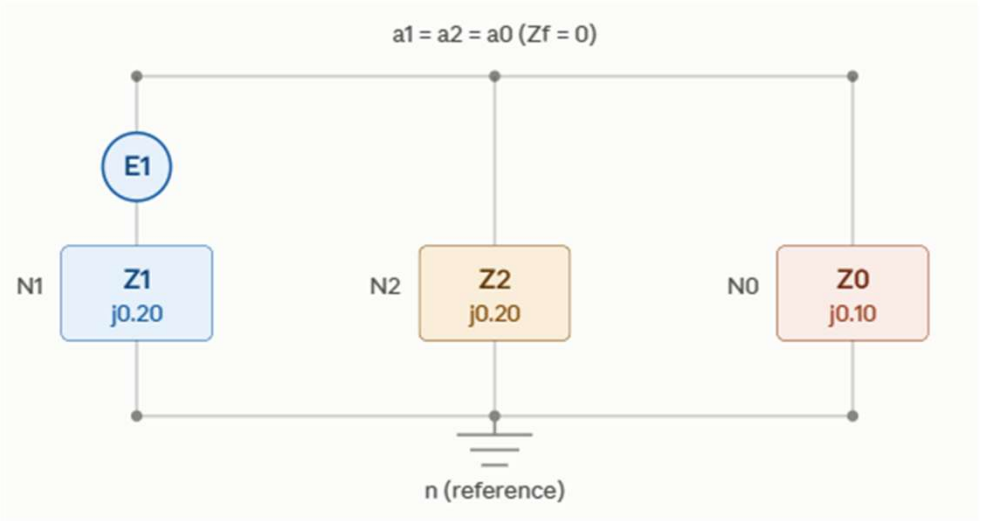
$$\boxed{V_1 = V_2}$$

Substituting back gives

$$\boxed{V_0 = V_1 = V_2}$$

So at the fault point,

$$\boxed{V_1 = V_2 = V_0}$$



For an L-L-G fault, all three sequence networks are required.

The negative- and zero-sequence impedances appear in parallel:

$$Z_2 \parallel Z_0 = \frac{Z_2 Z_0}{Z_2 + Z_0}$$

Therefore,

$$Z_2 \parallel Z_0 = \frac{(j0.20)(j0.10)}{j0.20 + j0.10}$$

$$= j0.0667$$

Hence,

$$I_1 = \frac{E_1}{Z_1 + (Z_2 \parallel Z_0)}$$

$$I_1 = \frac{1}{j0.20 + j0.0667}$$

$$\boxed{I_1 = -j3.75\ pu}$$

The sequence voltage at the fault is

$$V_1 = E_1 - Z_1 I_1$$

$$V_1 = 1 - (j0.20)(-j3.75)$$

$$\boxed{V_1 = 0.25\ pu}$$

For the negative-sequence network,

$$V_2 = -Z_2 I_2$$

At the fault,

$$V_2 = V_1 = 0.25$$

Therefore,

$$I_2 = -\frac{0.25}{j0.20}$$

$$\boxed{I_2 = -j1.25\ pu}$$

Similarly,

$$I_0 = -\frac{0.25}{j0.10}$$

$$\boxed{I_0 = -j2.50\ pu}$$

So,

$$\boxed{I_0 = -j2.50, \quad I_1 = -j3.75, \quad I_2 = -j1.25\ pu}$$

Use the transformation matrix:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} -j2.50 \\ -j3.75 \\ -j1.25 \end{bmatrix}$$

For phase a,

$$I_a = I_0 + I_1 + I_2$$

$$\boxed{I_a = -j7.50\ pu}$$

Worked Example 1: Three-Phase Symmetrical Fault

SYMMETRICAL FAULT PROBLEM

Problem: A 20 MVA, 10 kV alternator has a subtransient reactance of 5%. Find the additional series external reactance required so that the fault current on a 3-phase short circuit at the terminals does not exceed 8 times full-load current.

Step	Operation	Result
1	Base MVA = 20, Base kV = 10. Calculate base current: Ibase = 20×10^3 / ($\sqrt{3}$ × 10)	Ibase = **1154.7 A**
2	Maximum allowed fault current = 8 × IFL = 8 × 1154.7	If,max = **9237.6 A**
3	Required total pu impedance: Ztotal = 1.0 / (8 × 1.0) (since IFL = 1.0 pu)	Ztotal = **0.125 pu**
4	Alternator reactance already = 5% = 0.05 pu. External reactance needed: Xext = 0.125 − 0.05	Xext = **0.075 pu = 7.5%**
5	Convert to ohms: Xext(Ω) = 0.075 × (10^2/20)	Xext = **0.375 Ω**

Answer: An external series reactance of **0.375 Ω (7.5% on machine base)** limits the 3-phase fault current to exactly 8 × full-load current.